

KARTHICK SUBRAMANIAN MUTHUKKARUPPAN

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SUMMARY

MSc in AI candidate at National College of Ireland with 3 years of professional experience in quality engineering and automation testing at Cognizant Technology Solutions. Skilled in Selenium, Cypress, and CI/CD pipelines with hands-on expertise in functional, regression, and performance testing across mobile and web platforms. Experienced in Agile environments, defect tracking with Jira and TestRail, and cross-functional team collaboration. Currently exploring AI and machine learning applications, with a strong foundation in Python. Holds Stamp 2 permission to work in Ireland, with Stamp 1G eligibility after graduation.

TECHNICAL SKILLS

- **Testing and Automation:** Selenium WebDriver, Cypress, JMeter, Manual Testing, Functional Testing, Regression Testing, Smoke Testing, REST API Testing
- **Tools and Platforms:** Jira, TestRail, Jenkins, GitHub, CI/CD Pipelines, Agile/Scrum, Docker, Linux, Git
- **Languages:** Java, Python, JavaScript, C/C++, SQL, HTML, CSS
- **AI & Machine Learning:** Deep Learning, Generative AI, Explainable AI (XAI), Intelligent Agents, Decision Analytics, PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, Prompt Engineering
- **AI Security & Risk:** OWASP LLM Top 10, AI Red-Teaming (Garak, PyRIT, Promptfoo), Prompt Injection Testing, Threat Modelling, AI Governance (EU AI Act, NIST AI RMF)
- **Data, Cloud & Frameworks:** Pandas, NumPy, MySQL, MongoDB, AWS (S3, EC2), GCP (Vertex AI), Azure, FastAPI, Flask
- **Fundamentals:** Software Testing Lifecycle, Bug Lifecycle, Test Case Design, Database Validation

EXPERIENCE

AI Risk and Security Intern | RakFort Limited | Dublin, Ireland

July 2026 - Present

- Performed security evaluations of large language (LLM) applications by employing techniques such as a prompt injection, jailbreak and Data Loss Prevention (DLP).
- Performed red teaming on the applications leveraging tools like Garak, PyRIT, and Promptfoo.
- Assisted in producing AI Risk and Governance documentation, including model inventory, dataset provenance and risk register documents, all aligned to the EU AI Act and NIST AI RMF

Programmer Analyst | Cognizant Technology Solutions | Kochi, India

Oct 2023 - Jan 2026

- Performed **manual** and **automated testing** for the Paramount+ application across mobile and web platforms, covering functional, regression, and smoke testing.
- Supported automation testing using **Selenium** and **Cypress**, improving test coverage and enhancing testing efficiency across releases.
- Managed test cases, test execution, and defect tracking using **Jira** and **TestRail**, ensuring accurate documentation and effective issue resolution.
- Collaborated with developers, product managers, and QA teams in an **Agile** environment to identify and resolve critical defects ahead of release deadlines.

Programmer Analyst Trainee | Cognizant Technology Solutions | Kochi, India

Oct 2022 - Oct 2023

- Created and executed test cases to validate software functionality and ensure applications met quality standards under the guidance of senior QA engineers.
- Developed and maintained automated test scripts using **Selenium** and **Cypress** to improve test coverage and reduce manual effort.
- Identified, documented, and tracked software defects throughout the complete bug lifecycle until resolution using **Jira** and **TestRail**.
- Applied **Agile** testing methodologies and QA best practices in daily project work, contributing to sprint planning and release validation.

Quality Engineering and Assurance Intern | Cognizant Technology Solutions | India

Apr 2022 - Jul 2022

- Gained hands-on experience in **SQL** and **Selenium** through structured software testing activities and practical lab exercises.
- Developed practical skills in **HTML** and **CSS** while contributing to testing and interface validation tasks across team projects.

EDUCATION

MSc in Artificial Intelligence | National College of Ireland, Dublin

Jan 2026 - Jan 2027

Modules: Machine Learning, AI-Driven Decision Making, Engineering and Evaluating AI Systems, Intelligent Agents and Process Automation

First Class Honours (1:1) till Semester 1

BE in Electronics and Communication Engineering | VCET | Tamil Nadu, India

Aug 2018 - May 2022

First Class Honours (1:1)

CERTIFICATIONS AND ACHIEVEMENTS

- **Software Testing:** Masterclass Software Testing with Jira and Agile, Selenium Web Driver with Java, Cypress Automation Testing, Types of Testing and Bug Lifecycle, Web Driver IO with JavaScript Automation.
 - Developed **MindTune-OS**, an EEG-based agentic AI system using OAuth 2.0 and Groq, achieving 92.94% mental state classification accuracy with TinyML on Arduino | Link: <https://huggingface.co/spaces/Karthickmk/MindTune-OS/>
 - **Neuro-Symbolic XAI Pipeline**, Designed an explainable medical image diagnosis system integrating CNNs, symbolic rules, and SQLite-based prediction logging across chest & skin datasets | Link: <https://huggingface.co/spaces/Karthickmk/neuro-symbolic-medical-ai>
 - Built **Irish Culture Chat**, a React.js AI chatbot with interactive Maps API integration, delivering personalised cultural and heritage insights about Ireland | Link: <https://irish-chat.vercel.app/>
 - Created a decentralised rent deposit escrow for Ireland, "**Sol Rent**" powered by Solana. Deposits are locked in a PDA, move-in evidence is hashed on-chain, and release is instant with mutual-sign approval | Link: <https://solrent-app.vercel.app/>
 - Built an autonomous **Multi-Agent research system on Azure AI Foundry** (GPT-4o, LangChain, FastAPI, React) with an adaptive reasoning loop that searches real-time sources, detects information gaps and contradictions, and generates structured, confidence-scored research reports | Link: <https://research-intelligence-agent.onrender.com/>
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CONFERENCES AND RESEARCH

EuStats International Conference | Italy

May 2026

- Presented a research paper proposing a neuro-enhanced multi-agent architecture for autonomous smart agriculture, combining reinforcement learning, computer vision and IoT robotics to model future precision-farming systems.
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